

Methodology

The objective of this project was to predict whether a client would convert (1) or not convert (0) using a Logistic Regression model. As the competition evaluation metric is F1-score, the model is built to improve the balance between precision and recall rather than a higher accuracy score.

The dataset was first explored to understand the features, missing values, categorical variables, and class distribution. By observing the histograms, the distributions were observed. The target variable was found to be imbalanced, with non-converted clients forming the majority class. Missing values were present in variables such as Age(1480), Income(984), and Time_On_Site(1848).

At the feature selection step, User_ID and Campaign_Code were removed because they represent unique identifiers and do not contribute towards prediction.

At the feature engineering step, two new features were added. Ratio of product pages viewed to total pages viewed to capture browsing activity and Interaction between previous purchases and products viewed. The Time_On_Site variable showed a skewed distribution with a small number of extreme outliers. To reduce the effect of these outliers, the variable was capped at its 99th percentile value. This feature transformation improved the validation F1-score and was included in the final model at later stage.

The categorical variables Device_Type and Traffic_Source were transformed into numeric values using one-hot encoding.

Missing numerical values were imputed using the median of each feature. A small improvement in the F1-score was observed after “add_indicator=True” was added to preserve information about missingness.

All features were then standardized using StandardScaler to make sure that the variables were on the same scale before training the model.

A Logistic Regression model was trained using `class_weight='balanced'` to resolve the class imbalance problem.

This combination achieved the best validation F1-score among all other approaches that have been tested and was used to generate the final prediction on the `private_test.csv` dataset.